

ROCKHAMPTON, Australia

WMO: 943740

Lat: 23.3753S

Lon: 150.4775E

Elev: 15

StdP: 101.14

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	7.9	-6.1	2.3	21.8	-3.1	2.9	20.3	8.3	19.5	7.5	20.0	0.7	190	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	35.7	22.7	34.1	22.5	32.9	22.3	25.7	30.2	25.2	29.4	24.7	28.8	4.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	19.8	27.1	24.1	19.1	26.8	23.6	18.5	26.6	79.5	30.2	77.3	29.5	75.2	28.8	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	7.1	6.5	DB	3.9	39.1	1.0	1.5	3.2	40.2	2.6	41.0	2.1	41.9	1.4	43.0	(4)
(5)			WB	2.5	27.2	1.2	1.3	1.7	28.2	1.0	28.9	0.3	29.7	-0.6	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	27.3	26.9	26.0	23.7	20.4	17.9	17.0	18.2	21.2	23.7	25.4	26.8	(6)	
	DBStd	4.21	1.84	1.82	1.67	1.72	2.30	2.54	2.42	2.35	2.22	2.20	2.04	2.06	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	134	0	0	0	0	8	38	54	31	3	0	0	0	(9)	
	CDD10.0	4691	536	474	495	409	324	237	218	254	337	423	461	522	(10)	
	CDD18.3	1783	278	241	237	160	74	26	14	26	89	165	211	263	(11)	
	CDH23.3	15149	2746	2260	1959	1025	411	113	84	233	744	1260	1797	2517	(12)	
(13)	CDH26.7	5388	1088	848	669	253	50	7	3	36	213	453	713	1055	(13)	
(14)	Wind	WSAvg	3.3	3.8	3.8	3.6	3.4	2.9	2.7	2.6	2.8	3.0	3.4	3.5	3.7	(14)
(15) Precipitation	PrecAvg	811	113	132	102	44	49	36	28	25	27	49	70	114	(15)	
	PrecMax	1631	660	504	447	199	304	186	137	106	146	199	299	533	(16)	
	PrecMin	394	2	3	2	1	0	0	0	0	0	0	0	3	(17)	
	PrecStd	279	109	116	103	44	56	40	36	26	32	42	51	108	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	36.3	35.0	32.2	29.8	27.6	27.0	29.9	33.3	35.2	36.3	37.9	(19)	
		MCWB	23.8	24.1	22.9	20.8	19.1	18.1	17.1	17.1	18.3	20.0	20.8	22.4	(20)	
	2%	DB	35.0	34.1	32.8	30.2	28.0	25.9	25.3	27.4	30.9	32.8	33.9	35.1	(21)	
		MCWB	24.1	23.7	23.0	20.6	18.5	17.5	16.3	16.7	18.2	20.1	21.3	22.9	(22)	
	5%	DB	33.0	32.5	31.2	29.1	26.9	24.6	24.1	25.8	29.0	31.0	32.1	33.3	(23)	
		MCWB	23.5	23.4	22.6	20.2	18.1	17.0	15.8	16.1	17.6	20.2	21.1	22.6	(24)	
	10%	DB	31.4	31.1	30.1	28.1	25.8	23.3	23.0	24.2	27.4	29.1	30.6	31.8	(25)	
		MCWB	23.1	23.2	22.2	20.1	17.6	16.4	15.4	15.5	17.3	19.6	20.8	22.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.5	26.2	24.3	22.6	21.2	20.6	20.6	22.1	23.6	24.6	25.8	(27)	
		MCDB	31.8	30.9	30.1	27.6	25.7	23.7	23.5	25.4	26.8	29.6	29.5	30.9	(28)	
	2%	WB	25.7	25.6	25.2	23.3	21.3	19.8	19.3	19.4	21.0	22.6	23.6	25.1	(29)	
		MCDB	30.4	29.8	28.8	26.6	24.7	22.7	22.4	24.0	25.4	27.8	29.1	30.2	(30)	
	5%	WB	25.1	25.1	24.5	22.4	20.4	18.9	18.2	18.3	20.1	21.8	22.9	24.4	(31)	
		MCDB	29.3	29.0	28.0	26.1	24.1	22.4	21.3	22.7	25.1	26.9	28.5	29.3	(32)	
	10%	WB	24.6	24.5	23.9	21.7	19.5	18.1	17.2	17.4	19.4	21.1	22.4	23.8	(33)	
		MCDB	28.7	28.3	27.3	25.6	23.4	21.6	20.7	21.8	24.4	26.2	27.6	28.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	8.4	8.8	9.6	11.0	11.2	12.5	13.2	13.0	11.3	10.5	9.8	(35)	
		MCDBR	11.3	10.6	10.6	10.9	12.1	12.2	13.4	15.3	15.6	14.0	13.0	12.4	(36)	
	5% WB	MCWBR	3.2	3.1	3.3	3.6	4.5	5.2	6.0	6.2	5.5	4.5	3.8	3.6	(37)	
		MCWBR	9.4	8.8	8.4	8.7	9.1	9.7	12.3	12.4	11.6	11.0	10.2	10.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.395	0.387	0.371	0.351	0.327	0.318	0.308	0.320	0.354	0.380	0.395	0.398	(40)	
		taud	2.495	2.524	2.558	2.588	2.605	2.631	2.632	2.585	2.484	2.444	2.434	2.461	(41)	
	Ebn at Noon		949	943	932	909	894	883	905	928	934	939	943	948	(42)	
		Edh at Noon	116	111	102	92	83	77	79	90	108	119	123	120	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.58	6.16	5.55	4.87	4.17	3.61	4.07	4.91	5.96	6.60	6.90	6.87	(44)	
	RadStd		0.47	0.65	0.45	0.45	0.22	0.29	0.32	0.34	0.47	0.45	0.74	0.72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page